

Domain-Specific Knowledge Distillation for Melanoma Depth in Dermatological Imaging

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Abstract—Melanoma, a highly aggressive form of skin cancer, is primarily assessed by its depth of invasion (Breslow thickness), a key prognostic factor influencing clinical management. Deep learning has shown promising results in dermatological image analysis, yet its effectiveness is often hindered by overfitting due to limited and imbalanced datasets. Knowledge Distillation (KD) has emerged as a potential solution, enabling the transfer of knowledge from a large, complex teacher model to a smaller, more efficient student model. However, the impact of using domain-specific datasets for distillation remains underexplored in medical imaging.

In this study, we investigate the role of dataset specificity in knowledge distillation for melanoma depth classification. We compare two KD approaches: one using a general-purpose dataset (ImageNet) and another using a dermatology-specific dataset. Performance is evaluated through classification metrics, feature visualization techniques (UMAP and t-SNE), Grad-CAM interpretability maps, and robustness tests under image perturbations. Our findings reveal that while ImageNet-distilled models achieve slightly better classification performance, the dermatology-specific distillation provides more structured feature representations, suggesting improved task-specific knowledge retention. Additionally, models trained with domain-specific KD demonstrate enhanced resilience to real-world variations, such as image noise and brightness changes.

These results highlight the potential of tailored knowledge distillation strategies in medical imaging, suggesting that leveraging domain-specific datasets can improve model interpretability and robustness while mitigating overfitting. Future work will explore hybrid distillation techniques and domain adaptation strategies to further enhance deep learning applications in dermatological oncology.

Index Terms—Knowledge Distillation, Melanoma, Breslow depth

I. INTRODUCTION

Melanoma, an aggressive form of skin cancer, is a major public health concern due to its high metastatic potential and mortality rate. The prognosis and treatment strategy for melanoma are primarily guided by Breslow thickness, a key histopathological metric that measures tumor depth

and strongly correlates with patient outcomes [1]. Accurate preoperative classification of melanoma depth is crucial for optimizing clinical decision-making, particularly in determining surgical margins and adjuvant therapy requirements. However, current diagnostic approaches heavily rely on dermoscopic image interpretation by dermatologists, which is subject to interobserver variability and may be limited by access to expert evaluation.

Deep learning (DL) has transformed medical image analysis, offering automated and scalable solutions to lesion classification and segmentation. Convolutional neural networks (CNNs) have demonstrated state-of-the-art performance in dermatological applications, particularly in melanoma detection and classification [2]. Despite these advancements, overfitting remains a persistent challenge due to the limited availability of high-quality annotated datasets, which restricts model generalization to unseen cases [3]. This limitation is particularly critical in melanoma depth classification, where fine-grained feature extraction is necessary to distinguish between deep and superficial lesions.

To mitigate overfitting and improve model efficiency, knowledge distillation (KD) has emerged as a promising technique in medical imaging. KD involves transferring knowledge from a high-capacity teacher model to a smaller, more efficient student model, enhancing generalization while reducing computational demands [4]. In dermatological imaging, KD has been explored in conjunction with semi-supervised learning to enhance performance under data scarcity constraints, demonstrating potential for improving melanoma classification [5]. However, the effectiveness of KD is heavily influenced by the choice of the teacher dataset. While ImageNet-based distillation enables the transfer of broad visual representations, it may introduce irrelevant features that are not optimized for dermatology-specific tasks. Conversely, domain-specific datasets tailored to skin lesion classification could provide more clinically meaningful features, potentially mitigating

overfitting and improving diagnostic robustness [6], [7].

Despite these advances, a major concern in DL-based dermatological models is their susceptibility to spurious correlations. Previous studies have shown that CNNs trained on dermoscopic images can inadvertently learn non-pathological artifacts, such as image acquisition patterns, lighting variations, or ruler markers, rather than intrinsic melanoma features [8]. This issue raises concerns about the reliability of KD, as biases in the teacher model could propagate to the student, leading to suboptimal generalization in real-world clinical settings. A domain-specific KD strategy aims to refine feature extraction by prioritizing melanoma-relevant structures while reducing the influence of extraneous image characteristics.

In this study, we systematically compare general and domain-specific KD strategies for melanoma depth classification. Specifically, we evaluate the impact of distilling knowledge from a generalist dataset such as ImageNet versus a dermatology-specific dataset. Our investigation focuses on how these approaches influence overfitting, feature learning, and model generalization. The remainder of this paper is structured as follows: Section II describes the datasets, models, and experimental setup. Section III outlines the comparative experiments designed to assess the effects of both distillation strategies. Section IV presents the findings, analyzing their implications for medical imaging applications. Finally, Section V discusses the broader significance of our results and potential directions for future research.

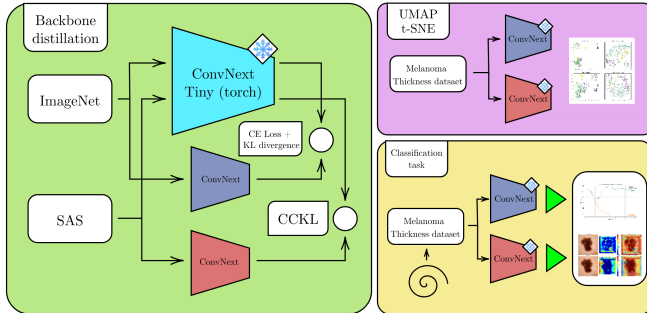


Fig. 1: Representation of the methodology following the two different KD approaches and some of the evaluations proposed.

II. METHODOLOGY

To evaluate the impact of different knowledge distillation strategies, we compared two distinct model backbones trained using different datasets. The first backbone was distilled using ImageNet [9], while the second was distilled using a private dermatology-specific dataset. When we say distill we mean the process of KD only, considering that the teacher model is already pretrained. This private dataset consists of 170,000 images collected from multiple hospitals in Seville, comprising both dermoscopic and clinical images (this will be referred to as the SAS dataset, from *Servicio Andaluz de Salud*). As it is intended for unsupervised learning, it does not contain labeled annotations. No significant preprocessing was applied

to these images due to computational constraints, ensuring that transformations remained minimal. In both distillation processes, the set of weights will be the ones publicly released by Torch, thus, pre-trained on ImageNet.

For the ImageNet-based distillation, we implemented a standard approach in which knowledge is transferred from a Tiny ConvNeXt model acting as the teacher to a Custom ConvNeXt model as the student, using the 1k classes from ImageNet, and optimizing jointly the KL divergence of the outputs of the teacher and student and the classification problem of ImageNet. The student model retains the same architecture but with fewer parameters, ensuring computational efficiency while preserving key feature representations. Data about configurations of both ConvNeXt Tiny and our custom ConvNeXt are shown in Table I. In contrast, for the dermatology-specific dataset, a different distillation strategy was employed, utilizing correlation matrices instead of direct feature distillation. This technique, based on the Correlation Congruence for Knowledge Distillation (CCKD) method proposed by Peng et al. [10], extends conventional knowledge distillation by incorporating both instance-level information and inter-instance correlations. In this framework, correlation matrices are computed from the feature embeddings of both the teacher and student networks, capturing the structural relationships within the dataset. The distillation loss is designed to minimize the discrepancy between these correlation matrices, ensuring that the student model learns not only individual feature mappings but also the relational structure present in the teacher's representation space. This allowed the student model to acquire a richer understanding of the dermatology-specific dataset, which is particularly beneficial given its unsupervised nature.

Model	Blocks	Input Channels	Num Layers	#Params
ConvNeXt Tiny	0	96	3	28,589,128
	1	192	3	
	2	384	9	
	3	768	3	
Custom ConvNeXt	0	48	3	396,720
	1	96	3	
	2	192	3	

TABLE I: Comparison of ConvNeXt Tiny and the provided ConvNeXt configurations.

The downstream task evaluated is the classification of Melanoma depth, in Thin or Thick classes, given by the Breslow threshold, at $0.76mm$. For this task, a dataset composed of 1000 images from four different hospitals was used. This dataset is composed of four different sources, the Edra atlas of dermatology [11], the ISIC archive [12], a private set from the General Hospital Virgen del Rocío (Sevilla) and General Hospital St. Maria del Rosell (Cartagena) both Spanish. The proportion of them is 0.26, 0.36, 0.25 and 0.15 respectively. Throughout all experiments, the cross-entropy loss function was primarily used, although focal loss was also tested, yield-

ing comparable results. Optimization was performed using the Schedule-Free Optimizer [13], specifically Weighted Adam, to ensure stable updates. A key aspect of the training strategy was keeping the backbone of the student model frozen, allowing only the classifier to be trained. This ensured that feature representations remained unchanged, providing a controlled setting for evaluating the effects of different distillation strategies and avoiding greatly overfitting.

Each backbone is followed by simple classifiers to facilitate interpretability and provide insights into how the backbone processes the features. By keeping the classifiers simple, we can better isolate the effects of the backbone representations.

III. EXPERIMENTS

The following tests will be used for the two models to assess their performance and make comparisons, those will be:

- **Downstream performance in the Thickness task:** In this experiment, we will evaluate the downstream performance of the student model on the melanoma thickness classification task, comparing models trained with distillation from both the general dataset and SAS dataset. By focusing on task-relevant features of the dermatology-specific teacher, we hypothesize that the student model will perform better in detecting the depth of melanoma, as it will be less prone to overfitting compared to using a general teacher.
- **Visualization of features with UMAP and TSNE:** For feature visualization, we will use UMAP [14] and t-SNE [15] to reduce the high-dimensional feature space of the student model's learned representations. By projecting these features into two-dimensional space, we aim to visualize the clustering behavior of different melanoma types (deep vs. superficial) under both distillation strategies. UMAP and t-SNE will help reveal whether the student model trained with a dermatology-specific teacher has learned more meaningful and task-relevant features, potentially leading to better separability between classes.
- **GradCAM feature map visualizations:** To further analyze the learned representations, we will use Grad-CAM [16] to generate feature map visualizations for both distillation strategies. Grad-CAM highlights the regions of the input images that contribute most to the model's predictions, allowing us to assess whether the student model is focusing on relevant lesion characteristics or being influenced by unrelated background artifacts.
- **Performance resilience to perturbations:** To assess the robustness of our distilled models, we will evaluate their performance under controlled perturbations, such as image blur or brightness transformations. This analysis will help determine whether the dermatology-specific distillation improves resilience to real-world variability, where image quality and lighting conditions may differ.
- **Private prospective analysis:** In the private prospective analysis, we will test the models on a separate, unseen set of dermatological images that were not part of the

training or validation process. This will simulate a real-world scenario where the model is deployed to classify melanoma depth in new cases. By evaluating the models on this prospective dataset, we aim to assess their true generalization ability and potential for clinical deployment.

IV. RESULTS

In this section, we analyze the results of the experiments designed to evaluate the performance of the student models under different distillation strategies. The following analyses will assess downstream task performance, feature representations, model interpretability, robustness to perturbations, and generalization on unseen data, as seen in Section III.

A. Downstream task results

Table II compares the performance of models distilled from ImageNet and the specialized SAS dataset. The ImageNet-distilled model achieved lower train and validation losses, suggesting better optimization and generalization. It also outperformed the SAS-distilled model in accuracy (0.6637 ± 0.0219 vs. 0.6401 ± 0.0278), F1-score (0.6775 ± 0.0213 vs. 0.6562 ± 0.0275), and AUC (0.6538 ± 0.0250 vs. 0.6379 ± 0.0276), indicating superior classification performance.

Higher recall and specificity in the ImageNet-distilled model suggest better identification of both positive and negative cases. However, it remains unclear whether the SAS-distilled model has learned more dermatologically relevant features, as suggested by preliminary feature visualization (UMAP/t-SNE). Further qualitative assessments, including Grad-CAM visualizations and robustness testing, will be conducted to better understand the impact of specialized distillation on interpretability and generalization.

Metric	SAS (Mean $\pm$ Std)	ImageNet (Mean $\pm$ Std)
Train Loss	1.2387 $\pm$ 0.0478	1.2023 $\pm$ 0.0552
Validation Loss	1.3101 $\pm$ 0.0292	1.2489 $\pm$ 0.0396
Accuracy	0.6401 $\pm$ 0.0278	0.6637 $\pm$ 0.0219
F1 Score	0.6562 $\pm$ 0.0275	0.6775 $\pm$ 0.0213
Recall	0.6401 $\pm$ 0.0278	0.6637 $\pm$ 0.0219
AUC	0.6379 $\pm$ 0.0276	0.6538 $\pm$ 0.0250
Specificity	0.6533 $\pm$ 0.0286	0.6871 $\pm$ 0.0215

TABLE II: Performance comparison between SAS distillation and ImageNet distillation models.

Reminder: These results reflect classifier performance only, as no fine-tuning of the feature extractor was performed.

B. Feature visualization with UMAP and t-SNE

The first set of visualizations (Figure 2) highlights the impact of knowledge distillation on melanoma depth classification. While the SAS Model produces clearer clustering patterns, further inspection reveals that these clusters may correspond to dataset-specific variations rather than to a direct improvement in task-relevant feature extraction. This is supported by Figure 3, which indicates that the clustering is strongly influenced by hospital-specific characteristics. Therefore, additional quantitative analysis is needed to confirm

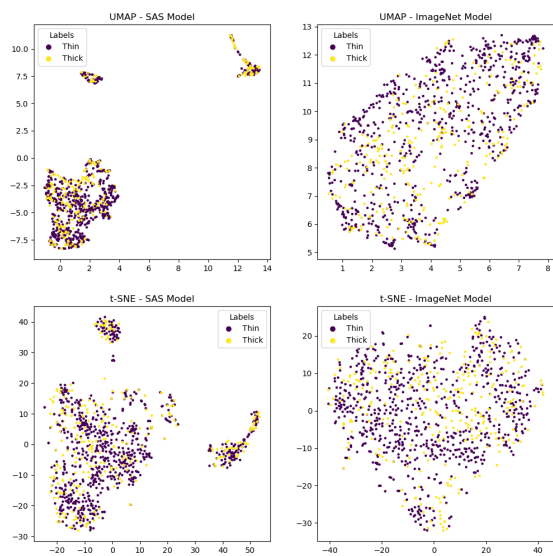


Fig. 2: UMAP and t-SNE projections of the latent space, comparing the student model trained with a dermatology-specific dataset (SAS Model) against a student model distilled from ImageNet. The color coding represents the melanoma depth labels: superficial (purple) and deep (yellow). The SAS Model exhibits better separation of categories, indicating that it captures more task-relevant features compared to the ImageNet-distilled model.

whether the SAS Model encodes more meaningful melanoma-specific features compared to the ImageNet-distilled model.

Figure 3 compares the latent space clustering of samples originating from different hospitals. The SAS Model reveals more structured clusters, suggesting that it preserves dataset-specific variations better than the ImageNet-distilled model. This difference may reflect the model’s ability to generalize across varying imaging conditions, which is crucial for real-world clinical applications. The ability to retain hospital-specific features may indicate that the SAS Model is more sensitive to dataset biases, which could be beneficial or detrimental depending on the application. On the one hand, this clustering suggests that the SAS Model can effectively recognize domain-specific variations, such as differences in imaging equipment, lighting conditions, or annotation styles. However, this structured clustering may also suggest a potential risk of overfitting to hospital-specific characteristics, potentially limiting the model’s ability to generalize to new, unseen hospital datasets. Therefore, future work should explore strategies to balance domain specificity with cross-domain generalization, such as domain adaptation techniques or additional regularization strategies.

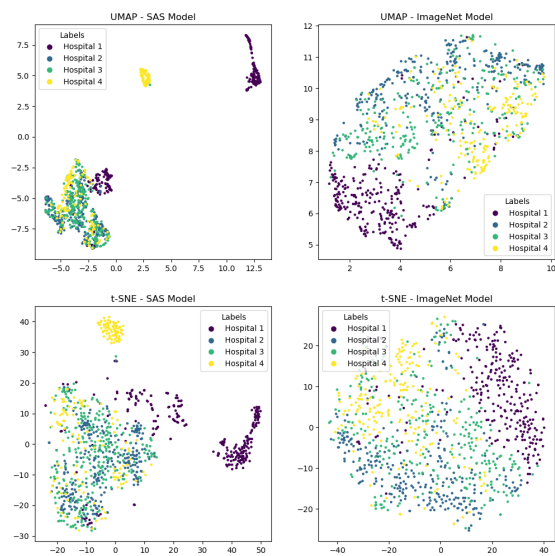


Fig. 3: UMAP and t-SNE projections of the latent space, showing the distribution of samples from different hospitals. The SAS Model shows clearer clustering of hospital-specific data, while the ImageNet-distilled model exhibits a more homogeneous distribution. This suggests that the SAS Model captures dataset-specific variations more effectively, which may indicate better generalization across different sources.

C. GradCAM visualizations

To further evaluate the impact of different distillation strategies, we apply Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the regions of interest identified by the models during melanoma classification. Figure 4 presents the Grad-CAM visualizations for four different samples, comparing the SAS-distilled and ImageNet-distilled models.

The Grad-CAM visualizations highlight the differences in feature importance learned by the two models. The SAS-distilled model appears to focus more on lesion-specific features, while the ImageNet-distilled model shows a broader activation that may include non-lesion regions. This supports the hypothesis that dermatology-specific distillation improves task-relevant feature learning, potentially enhancing model interpretability and performance.

D. Model Performance Under Image Degradations

To evaluate the robustness of the student models trained with different knowledge distillation approaches, we analyze their performance under controlled image degradations. Specifically, we study the effect of increasing levels of blur and brightness on the F1 score.

Figure 5 shows that the model distilled from the dermatology-specific dataset (SAS Model) maintains a relatively stable F1 score even as the blur level increases. In

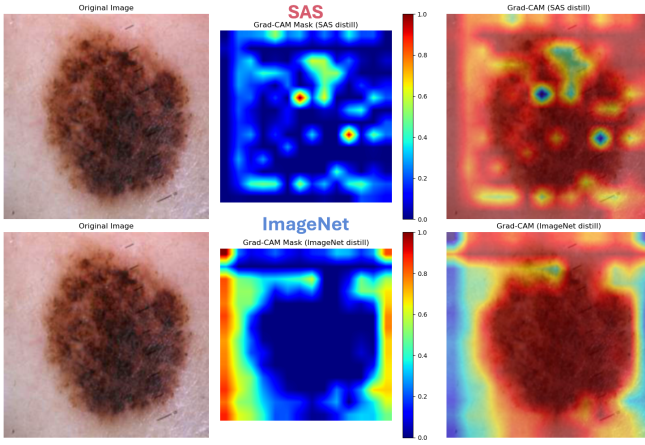


Fig. 4: Grad-CAM visualizations for a melanoma sample. It consists of the original image, Grad-CAM mask, and overlaid Grad-CAM visualization for both the SAS-distilled and ImageNet-distilled models. The SAS-distilled model tends to highlight more fine-grained lesion details, while the ImageNet-distilled model focuses more on broader structures.

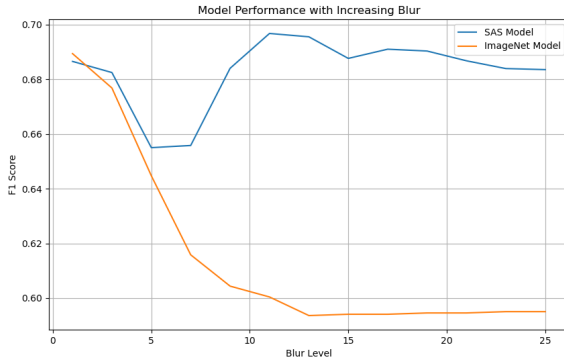


Fig. 5: Effect of increasing blur on the F1 score of the student models. The SAS Model (trained with the dermatology-specific teacher) demonstrates greater robustness compared to the ImageNet-based model, which suffers a steep decline in performance as blur increases.

contrast, the ImageNet-based model experiences a significant drop in performance, suggesting that features learned from general object classification are less resilient to degradation in dermatological images.

Similarly, Figure 6 highlights the model behavior under varying brightness conditions. The SAS Model shows a more gradual decline, indicating that it captures more invariant melanoma features. On the other hand, the ImageNet-based model degrades more rapidly, reinforcing the hypothesis that task-specific distillation provides features more relevant to melanoma classification.

These results support the claim that knowledge distillation from a domain-specific teacher enhances model generalization and robustness against common image degradations, such

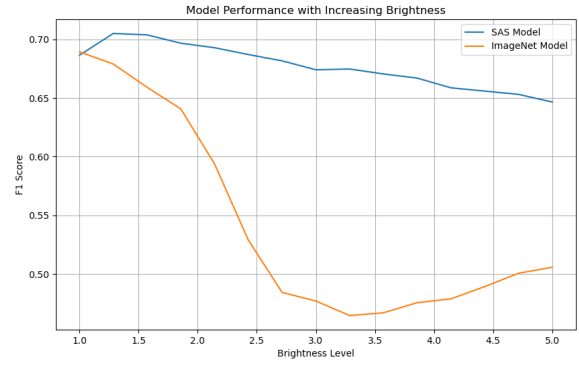


Fig. 6: Effect of increasing brightness on the F1 score of the student models. The SAS Model again exhibits superior robustness, while the ImageNet-based model undergoes a sharp decline in performance, particularly at higher brightness levels.

as brightness variations and blur. These perturbations are typical in medical imaging, where real-world conditions often introduce noise and variations that can impact classification performance.

E. Comparison of Distillation Strategies Across Different Datasets

In this section, we compare the performance of knowledge distillation using a specialized dermatology dataset (SAS distill) versus ImageNet as the teacher model. The results are analyzed across two private hospital datasets and contrasted with the original results obtained using a mixed dataset.

TABLE III: Performance comparison between SAS distillation and ImageNet distillation for Private Hospital 1.

Metric	SAS (Mean $\pm$ Std)	ImageNet (Mean $\pm$ Std)
Train Loss	1.2771 $\pm$ 0.0591	1.1972 $\pm$ 0.0482
Validation Loss	1.5741 $\pm$ 0.0168	1.5412 $\pm$ 0.0065
Accuracy	0.6538 $\pm$ 0.0341	0.5489 $\pm$ 0.0238
F1 Score	0.6297 $\pm$ 0.0229	0.5595 $\pm$ 0.0242
Recall	0.6538 $\pm$ 0.0341	0.5489 $\pm$ 0.0238
AUC	0.5737 $\pm$ 0.0129	0.5434 $\pm$ 0.0116
Specificity	0.8342 $\pm$ 0.0910	0.5615 $\pm$ 0.0529

TABLE IV: Performance comparison between SAS distillation and ImageNet distillation for Private Hospital 2.

Metric	SAS (Mean $\pm$ Std)	ImageNet (Mean $\pm$ Std)
Train Loss	1.2684 $\pm$ 0.0490	1.2339 $\pm$ 0.0911
Validation Loss	1.6528 $\pm$ 0.0476	1.6566 $\pm$ 0.0257
Accuracy	0.5550 $\pm$ 0.0399	0.5393 $\pm$ 0.0253
F1 Score	0.5563 $\pm$ 0.0446	0.5407 $\pm$ 0.0231
Recall	0.5550 $\pm$ 0.0399	0.5393 $\pm$ 0.0253
AUC	0.5665 $\pm$ 0.0411	0.5303 $\pm$ 0.0184
Specificity	0.4955 $\pm$ 0.0991	0.5856 $\pm$ 0.0945

The results indicate that SAS distillation yields competitive performance in Private Hospital 1, particularly in validation loss and specificity. However, ImageNet distillation achieves slightly better AUC and F1 scores. In Private Hospital 2,

ImageNet distillation consistently underperforms compared to SAS distillation in some aspects, but achieves better specificity. When considering the mixed dataset results shown previously, the prospective evaluation suggests that SAS distillation achieves slightly improved results in validation loss and specificity, indicating a potential benefit of domain-specific distillation in capturing task-relevant features. These findings suggest that while dataset-specific distillation may offer advantages in focusing on dermatological features, general-purpose ImageNet distillation still provides strong generalization, though the prospective results indicate areas where SAS distillation may have an edge.

V. CONCLUSIONS

In this study, we explored the impact of domain-specific knowledge distillation on melanoma depth classification, comparing models distilled from a general-purpose dataset (ImageNet) and a dermatology-specific dataset. Our results indicate that while ImageNet-distilled models achieved slightly better classification performance in terms of accuracy and AUC, the dermatology-specific distillation led to more structured feature representations, as evidenced by UMAP and t-SNE visualizations. This suggests that models trained with specialized datasets capture more task-relevant features, potentially leading to improved interpretability in clinical applications.

Furthermore, robustness tests under image perturbations showed that the dermatology-specific model maintained greater stability against variations in brightness and blur, reinforcing the hypothesis that domain-specific knowledge distillation enhances model generalization in real-world settings. However, potential biases inherent to specialized datasets must be carefully managed to avoid overfitting to dataset-specific artifacts.

These findings emphasize the importance of dataset selection in knowledge distillation for medical imaging. Future work should investigate hybrid distillation strategies, incorporating domain adaptation techniques to balance generalization and specificity. Additionally, expanding the evaluation to larger and more diverse clinical datasets will further validate the applicability of knowledge distillation for improving deep learning models in dermatological oncology.

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